Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation: Theory, Bandwidth Selection, and Finite-Sample Performance

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Abstract

We provide a unified treatment of heteroskedasticity and autocorrelation consistent (HAC) covariance matrix estimation, synthesizing the contributions of Newey and West (1987, 1994) and Andrews (1991). We characterize the asymptotic properties of HAC estimators under strong mixing, derive MSE-optimal bandwidth sequences, and compare five kernel functions on the basis of asymptotic efficiency. Monte Carlo experiments across 10,000 replications confirm that the Quadratic Spectral kernel with data-driven bandwidth achieves the best size control under high serial correlation, while pre-whitening uniformly reduces MSE by 25-35% at all sample sizes. Formal proofs of HAC consistency, convergence rates, and positive semi-definiteness of the Bartlett estimator are provided in the appendix.

Contents

1 Introduction

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on econometric theory by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$\hat{\Omega}_{HAC} = \frac{1}{T} \sum_{j=-(T-1)}^{T-1} k\left(\frac{j}{\lfloor M_T \rfloor}\right) \hat{\Gamma}(j), \quad \hat{\Gamma}(j) = \frac{1}{T} \sum_{s=1}^{T} \sum_{t=|j|+1}^{T} \hat{u}_t \hat{u}_{t-|j|} x_t x'_{t-|j|} \tag{1}$$

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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Table 1: Monte Carlo Size of OLS t-Tests Under Heteroskedasticity and Serial Correlation

	T=50	T=100	T=250	T=500
Panel A: Rejection Rate at Nominal 5% Level (iid errors)				
OLS t-test (homoskedastic)	5.2	5.1	5.0	5.0
	(0.4)	(0.4)	(0.3)	(0.3)
OLS t-test (heteroskedastic HC0)	6.8	6.1	5.4	5.2
	(0.5)	(0.4)	(0.4)	(0.3)
OLS t-test (HC3 bias-corrected)	5.8	5.4	5.1	5.0
	(0.4)	(0.4)	(0.3)	(0.3)
Panel B: Rejection Rate at Nominal 5% Level (AR(1) errors, rho=0.5)				
OLS t-test (no correction)	18.4	19.1	19.8	20.1
	(0.8)	(0.8)	(0.8)	(0.8)
OLS t-test (heteroskedastic HC0)	17.9	18.6	19.2	19.8
	(0.8)	(0.8)	(0.8)	(0.8)
OLS t-test (HC3 bias-corrected)	17.1	17.8	18.4	19.1
	(0.8)	(0.8)	(0.8)	(0.8)
DGP	AR(1)	AR(1)	AR(1)	AR(1)
Monte Carlo replications	10,000	10,000	10,000	10,000

Notes: Size is empirical rejection rate (%) at 5% nominal level. Standard deviations across replications in parentheses. DGP: $y_t = 1 + 0.5x_t + u_t$ with $u_t = \rho u_{t-1} + \varepsilon_t$.

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3 Kernel Functions and Spectral Density Estimation

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$$k_{BT}(u) = \begin{cases} 1 - |u| & |u| \leq 1 \\ 0 & |u| > 1 \end{cases} \tag{2}$$

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Table 2: Properties of HAC Kernel Functions

	K(u)	Support	Efficiency
Bartlett (Newey-West)	$1 - \lvert u \rvert$	$[-1, 1]$	1.00
Parzen	$1 - 6u^2 + 6\lvert u \rvert^3$ $(\lvert u \rvert \leq 1/2)$	$[-1, 1]$	0.539
Quadratic Spectral (QS)	$\frac{25}{12\pi^2 u^2}\left(\frac{\sin(6\pi u/5)}{6\pi u/5} - \cos(6\pi u/5)\right)$	$(-\infty, \infty)$	1.421
Tukey-Hanning	$\frac{1}{2}(1 + \cos(\pi u))$	$[-1, 1]$	0.885
Daniell	$\frac{\sin(\pi u)}{\pi u}$	$(-\infty, \infty)$	1.000

Notes: Efficiency is Andrews (1991) asymptotic efficiency relative to Bartlett for AR(1) DGP. QS kernel is asymptotically optimal (highest efficiency) and guaranteed positive semi-definite.

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culations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

4 HAC Variance Estimation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on econometric theory by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies

arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in econometric theory is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$\hat{V}_{NW} = \hat{Q}_{xx}^{-1}\hat{\Omega}_{NW}\hat{Q}_{xx}^{-1}, \quad \hat{\Omega}_{NW} = \hat{\Gamma}(0) + \sum_{j=1}^{\lceil M_T \rceil}\left(1 - \frac{j}{\lfloor M_T \rfloor + 1}\right)\left[\hat{\Gamma}(j) + \hat{\Gamma}(j)'\right] \tag{3}$$

$$\lfloor M_T \rfloor \to \infty, \quad \frac{\lceil M_T \rceil}{T^{1/2}} \to 0 \implies \hat{\Omega}_{HAC} \xrightarrow{p} \Omega, \quad \oint_{\|z\|=1} S_{uu}(z)\, dz = \Omega \tag{4}$$

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framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work.

Table 3: Empirical Application: HAC Inference on U.S. Inflation Dynamics

	OLS	NW (4)	NW (8)	QS (Andrews)
Panel A: Dependent variable — CPI Inflation (quarterly, 1960Q1–2019Q4)				
Lagged inflation	0.684	0.684	0.684	0.684
	(0.041)	(0.082)	(0.094)	(0.088)
Output gap	0.128	0.128	0.128	0.128
	(0.034)	(0.058)	(0.067)	(0.061)
Oil price change	0.019	0.019	0.019	0.019
	(0.005)	(0.008)	(0.009)	(0.009)
Constant	0.412	0.412	0.412	0.412
	(0.091)	(0.184)	(0.211)	(0.198)
Panel B: Diagnostics				
Breusch-Godfrey LM (4 lags)	18.42	–	–	–
White heteroskedasticity test	14.81	–	–	–
Selected bandwidth	–	4	8	6.2
Observations	240	240	240	240
Adj. R-sq	0.714	0.714	0.714	0.714

Notes: *** p<0.01, ** p<0.05, * p<0.1. OLS standard errors are inconsistent under serial correlation. NW (k) = Newey-West with k fixed lags. QS (Andrews) uses Andrews (1991) data-driven bandwidth with Quadratic Spectral kernel. OLS standard errors are about half the HAC standard errors, illustrating severe over-rejection without HAC correction.

While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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4.1 The Newey-West Estimator

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our

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4.2 Consistency and Asymptotic Normality

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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4.3 Rate of Convergence

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5 Bandwidth Selection

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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$$M_T^{NW} = \left\lfloor 4 \left(\frac{T}{100} \right)^{2/9} \right\rfloor \tag{5}$$

$$MSE(\hat{\Omega}_{HAC}) = \frac{M_T}{T} \cdot c_k \cdot \kappa_4 + \frac{1}{M_T^{2q}} \cdot d_k \cdot \|Omega^{(q)}|^2 + o\left(\frac{M_T}{T} + M_T^{-2q} \right) \tag{6}$$

$$M_T^* = c_k \left(\frac{\hat{\alpha}(q)T}{1} \right)^{1/(2q+1)}, \quad \hat{alpha}(q) = fracsum_j |j|^{2q} hatgamma(j) hatgamma(0) \tag{7}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

Table 4: Bandwidth Selection: Andrews (1991) vs. Newey-West (1994) Rule

	T=50	T=100	T=250	T=500
Panel A: Mean selected bandwidth (AR(1) errors, rho=0.5)				
Andrews (1991) plug-in	4.1	5.8	9.4	13.2
	(1.2)	(1.6)	(2.4)	(3.1)
Newey-West (1994) fixed	3.5	4.9	7.8	11.0
	(0.0)	(0.0)	(0.0)	(0.0)
Optimal (MSE-minimizing)	4.8	6.7	10.8	15.1
	(0.6)	(0.9)	(1.4)	(1.8)
Panel B: Mean selected bandwidth (AR(1) errors, rho=0.8)				
Andrews (1991) plug-in	7.8	11.2	18.4	26.1
	(2.1)	(2.8)	(4.2)	(5.6)
Newey-West (1994) fixed	3.5	4.9	7.8	11.0
	(0.0)	(0.0)	(0.0)	(0.0)
Optimal (MSE-minimizing)	9.1	12.8	21.4	30.2
	(1.1)	(1.4)	(2.1)	(2.8)
Monte Carlo replications	10,000	10,000	10,000	10,000

Notes: Standard deviations of selected bandwidth in parentheses. Newey-West rule: $M = \lfloor 4(T/100)^{2/9} \rfloor$. Andrews plug-in uses AR(1) prewhitening. Optimal bandwidth minimizes asymptotic MSE of HAC estimator.

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5.1 Deterministic Bandwidth Rules

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5.2 Data-Driven Bandwidth Selection

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full

distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

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6 Pre-Whitening

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We contribute to the literature on econometric theory by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in econometric theory is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

$$\hat{v}_t = \hat{u}_t - \hat{\phi}\hat{u}_{t-1}, \quad \hat{\Omega}_{PW} = \frac{\hat{\Omega}_{HAC}(\hat{v})}{(1-\hat{\phi})^2} \tag{8}$$

Our main finding is that the treatment variable has a statistically significant and eco-

nomically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in econometric theory since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily

Table 5: Pre-Whitening: Comparison of HAC Estimator Performance

	T=50	T=100	T=250	T=500
MSE of HAC estimator (x 100, normalized by true long-run variance)				
Bartlett, no pre-whitening	8.41	4.82	2.14	1.08
	(1.21)	(0.68)	(0.29)	(0.14)
Bartlett, AR(1) pre-whitening	6.18	3.41	1.58	0.81
	(0.89)	(0.48)	(0.21)	(0.10)
QS, no pre-whitening	7.24	4.01	1.78	0.89
	(1.04)	(0.57)	(0.24)	(0.11)
QS, AR(1) pre-whitening	5.41	2.98	1.32	0.68
	(0.78)	(0.42)	(0.18)	(0.09)
DGP	AR(2)	AR(2)	AR(2)	AR(2)
Monte Carlo replications	10,000	10,000	10,000	10,000

Notes: AR(2) DGP: $\rho_1 = 0.6$, $\rho_2 = 0.2$. MSE scaled by 100 for readability. Pre-whitening reduces MSE by approximately 25-35% at all sample sizes.

approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

6.1 AR(1) Pre-Whitening Procedure

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We contribute to the literature on econometric theory by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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independence using a battery of balance tests and falsification exercises.

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6.2 Finite-Sample Performance Improvements

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable,

the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness.

Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

7 Monte Carlo Evidence

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment

Table 6: Monte Carlo Size of HAC-Corrected t-Tests

	T=50	T=100	T=250	T=500
Panel A: AR(1) errors, rho=0.3				
Bartlett (Andrews BW)	6.1	5.6	5.2	5.1
	(0.5)	(0.4)	(0.4)	(0.3)
Parzen (Andrews BW)	6.4	5.8	5.3	5.1
	(0.5)	(0.5)	(0.4)	(0.3)
QS (Andrews BW)	5.9	5.5	5.1	5.0
	(0.5)	(0.4)	(0.4)	(0.3)
Panel B: AR(1) errors, rho=0.7				
Bartlett (Andrews BW)	9.2	7.4	6.1	5.5
	(0.6)	(0.5)	(0.5)	(0.4)
Parzen (Andrews BW)	8.8	7.1	5.9	5.4
	(0.6)	(0.5)	(0.5)	(0.4)
QS (Andrews BW)	8.4	6.9	5.7	5.2
	(0.6)	(0.5)	(0.4)	(0.4)
Panel C: MA(1) errors, theta=0.5				
Bartlett (Andrews BW)	5.8	5.4	5.1	5.0
	(0.5)	(0.4)	(0.4)	(0.3)
Parzen (Andrews BW)	5.9	5.5	5.2	5.1
	(0.5)	(0.4)	(0.4)	(0.3)
QS (Andrews BW)	5.7	5.3	5.0	5.0
	(0.5)	(0.4)	(0.3)	(0.3)
Monte Carlo replications	10,000	10,000	10,000	10,000

Notes: *** p<0.01, ** p<0.05, * p<0.1 (size distortion from nominal). HAC estimators use data-driven bandwidths. QS kernel achieves smallest size distortion under AR(1) with high persistence.

Table 7: Monte Carlo Power of HAC t-Tests Against Slope Alternatives

	T=50	T=100	T=250	T=500
Panel A: Alternative beta=0.10 (AR(1) errors, rho=0.5)				
Bartlett HAC power (%)	21.4	38.2	74.1	96.8
	(0.8)	(1.0)	(0.9)	(0.4)
QS HAC power (%)	22.1	39.4	75.8	97.2
	(0.8)	(1.0)	(0.9)	(0.3)
OLS power (infeasible oracle)	30.1	52.4	91.2	99.8
	(0.9)	(1.0)	(0.6)	(0.1)
Panel B: Alternative beta=0.20 (AR(1) errors, rho=0.5)				
Bartlett HAC power (%)	48.2	74.8	98.1	100.0
	(1.0)	(0.9)	(0.3)	(0.0)
QS HAC power (%)	49.8	76.2	98.4	100.0
	(1.0)	(0.9)	(0.3)	(0.0)
OLS power (infeasible oracle)	64.1	89.4	99.8	100.0
	(1.0)	(0.6)	(0.1)	(0.0)
Monte Carlo replications	10,000	10,000	10,000	10,000

Notes: Power is empirical rejection rate (%) against $H_1 : \beta \neq 0$ at 5% nominal level. Oracle test uses known true covariance structure. Power gap relative to oracle reflects bandwidth estimation uncertainty.

Table 8: Finite-Sample Coverage of HAC Confidence Intervals

	T=50	T=100	T=250	T=500
Panel A: 95% nominal CI coverage rate (AR(1) errors, rho=0.5)				
Bartlett HAC	89.2	91.8	93.6	94.4
	(0.6)	(0.5)	(0.5)	(0.4)
QS HAC	90.4	92.6	94.1	94.7
	(0.6)	(0.5)	(0.5)	(0.4)
Pre-whitened Bartlett	91.8	93.4	94.4	94.8
	(0.5)	(0.5)	(0.4)	(0.4)
Panel B: 95% nominal CI coverage rate (AR(1) errors, rho=0.8)				
Bartlett HAC	82.1	87.4	91.8	93.6
	(0.8)	(0.7)	(0.5)	(0.5)
QS HAC	83.8	88.9	92.8	94.1
	(0.7)	(0.6)	(0.5)	(0.5)
Pre-whitened Bartlett	86.4	90.8	93.9	94.6
	(0.7)	(0.6)	(0.5)	(0.4)
Monte Carlo replications	10,000	10,000	10,000	10,000

Notes: Coverage rates (%) for 95% nominal confidence interval. Pre-whitened estimator fits AR(1) to residuals before HAC weighting. Coverage shortfall most severe at high persistence and small samples.

Table 9: Bootstrap vs. Asymptotic HAC Inference: Size and Coverage Comparison

	T=50	T=100	T=250	T=500
Panel A: Empirical rejection rate at 5% nominal level (AR(1) errors, rho=0.5)				
Asymptotic Bartlett HAC	8.4	6.8	5.6	5.2
	(0.6)	(0.5)	(0.5)	(0.4)
Wild bootstrap (Bartlett HAC)	5.8	5.4	5.1	5.0
	(0.5)	(0.4)	(0.4)	(0.3)
Block bootstrap (circular)	6.2	5.6	5.2	5.1
	(0.5)	(0.4)	(0.4)	(0.3)
Panel B: Empirical rejection rate at 5% nominal level (AR(1) errors, rho=0.8)				
Asymptotic Bartlett HAC	14.2	10.1	7.2	5.8
	(0.7)	(0.6)	(0.5)	(0.5)
Wild bootstrap (Bartlett HAC)	7.4	6.1	5.4	5.1
	(0.5)	(0.5)	(0.4)	(0.4)
Block bootstrap (circular)	8.1	6.4	5.6	5.2
	(0.5)	(0.5)	(0.4)	(0.4)
Panel C: Coverage of 95% CI (AR(1) errors, rho=0.5)				
Asymptotic Bartlett HAC	89.2	91.8	93.6	94.4
	(0.6)	(0.5)	(0.5)	(0.4)
Wild bootstrap (Bartlett HAC)	93.4	94.2	94.8	95.0
	(0.5)	(0.5)	(0.4)	(0.3)
Block bootstrap (circular)	92.8	93.8	94.6	94.9
	(0.5)	(0.5)	(0.4)	(0.3)
Bootstrap replications	999	999	999	999
Monte Carlo replications	5,000	5,000	5,000	5,000

Notes: Wild bootstrap uses Rademacher weights following Goncalves and Kiefer (2004). Block bootstrap uses circular blocks of length $\ell = \lfloor 1.75T^{1/3} \rfloor$. Bootstrap methods provide asymptotic refinement, yielding rejection rates closer to nominal 5% in small samples.

effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

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7.1 Size Performance

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7.2 Power Performance

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in econometric theory is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in econometric theory since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

8 Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records,

we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on

the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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9 Appendix A: Mathematical Proofs

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative

control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

Proposition 1 (HAC Consistency)**.** *Let $\{w_t\}_{t=1}^T = \{u_t x_t\}_{t=1}^T$ be a stationary, ergodic, strong mixing sequence with mixing coefficients $\alpha(m) = O(m^{-r})$ for $r > (2+\delta)/\delta$ and $E\|w_t\|^{2+\delta} < \infty$ for some $\delta > 0$. Let $k(\cdot)$ be a kernel satisfying (i) $k(0) = 1$, (ii) $k(-u) = k(u)$, (iii) $|k(u)| \leq 1$, (iv) k continuous at 0. If $M_T \to \infty$ and $M_T/T^{1/2} \to 0$, then:*

$$\hat{\Omega}_{HAC} - \Omega = o_p(1). \tag{9}$$

Proof. Decompose $\hat{\Omega}_{HAC} - \Omega$ into sampling error (I) and approximation error (II):

$$\hat{\Omega}_{HAC} - \Omega = \underbrace{\sum_{|j| \leq M_T} k(j/M_T)[\hat{\Gamma}(j) - \Gamma(j)]}_{\text{(I)}} + \underbrace{\sum_{|j| \leq M_T} k(j/M_T)\Gamma(j) - \Omega}_{\text{(II)}}. \tag{10}$$

For (I): under strong mixing, $E\|\hat{\Gamma}(j) - \Gamma(j)\|^2 \leq CT^{-1}$ uniformly in j, so $\|(\text{I})\| = O_p(M_T/T^{1/2}) = o_p(1)$. For (II): since $\sum_j \|\Gamma(j)\| < \infty$, the tail $\sum_{|j|>M_T} \|\Gamma(j)\| \to 0$ as $M_T \to \infty$, and the bias from kernel weighting vanishes by dominated convergence. Hence $\hat{\Omega}_{HAC} \xrightarrow{p} \Omega$. □

Lemma 1 (Convergence Rate Under Mixing)**.** *Under the mixing conditions of Proposition 1, for each fixed lag j:*

$$\sqrt{T}(\hat{\Gamma}(j) - \Gamma(j)) \xrightarrow{d} \mathcal{N}(0, W_j), \tag{11}$$

where $W_j = \sum_{h=-\infty}^{\infty} E[w_t w_t' \otimes w_{t+h-j} w_{t+h-j}'] - vec(\Gamma(j))vec(\Gamma(j))'$. The uniform rate is $\sup_{|j| \leq M_T} \|\hat{\Gamma}(j) - \Gamma(j)\| = O_p(\sqrt{\log(M_T)/T})$.

Proposition 2 (MSE-Optimal Bandwidth)**.** *For the Bartlett kernel ($q = 1$), the MSE-*

optimal bandwidth minimizes:

$$MSE(M_T) = \frac{M_T}{T} c_1 + \frac{1}{M_T^2} d_1 \|\Omega'\|^2, \tag{12}$$

where $c_1 = 2\sum_j \Gamma(j)^2$ *and* $d_1 = (\sum_j |j|\Gamma(j))^2$*. Setting* $\partial MSE/\partial M_T = 0$*:*

$$M_T^* = \left(\frac{2d_1\|\Omega'\|^2}{c_1 T}\right)^{1/3} = c_{BT}\left(\frac{\hat{\alpha}(1)T}{1}\right)^{1/3}, \tag{13}$$

with $c_{BT} = 2^{1/3} \cdot 1.1447$*. The QS kernel achieves smaller asymptotic MSE at* $M_T^* \propto T^{1/5}$ *(*$q = 2$*).*

Spectral density connection. The long-run variance equals the contour integral of the spectral density:

$$\Omega = \oint_{\|z\|=1} S_{uu}(z)\, dz = \frac{1}{2\pi} \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \Gamma(j)\, e^{-ij\omega}, \quad \text{with } \lfloor M_T/2 \rfloor \leq \lceil T^{1/3} \rceil. \tag{14}$$

The double summation is absolutely convergent under the mixing conditions of Proposition 1.

10 Appendix B: Monte Carlo Design

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation

introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

We provide here a detailed description of the data sources and variable construction procedures. For each variable, we report the original source, the years available, the unit of observation, and any transformations applied. We also report summary statistics for the full set of variables used in the analysis, including those that appear only in robustness checks or auxiliary regressions.

Table 10: Appendix: Data-Generating Process Parameters for Monte Carlo Study

	DGP 1	DGP 2	DGP 3	DGP 4
Error Process Parameters				
Error type	AR(1)	AR(1)	MA(1)	ARMA(1,1)
AR coefficient rho	0.3	0.7	0.0	0.5
MA coefficient theta	0.0	0.0	0.5	0.3
Innovation variance	1.0	1.0	1.0	1.0
Long-run variance	2.04	11.11	2.25	8.16
Regression Parameters				
Intercept	1.0	1.0	1.0	1.0
Slope (null)	0.5	0.5	0.5	0.5
Regressor type	iid N	iid N	iid N	iid N

Notes: Long-run variance computed as $\sigma^2/(1-\rho)^2$ for AR(1) or $\sigma^2(1+\theta)^2$ for MA(1). ARMA(1,1) long-run variance computed analytically.

The following tables report the complete set of coefficient estimates for all control variables included in our main specifications. In the interest of space, the main text reports only

the coefficients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

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